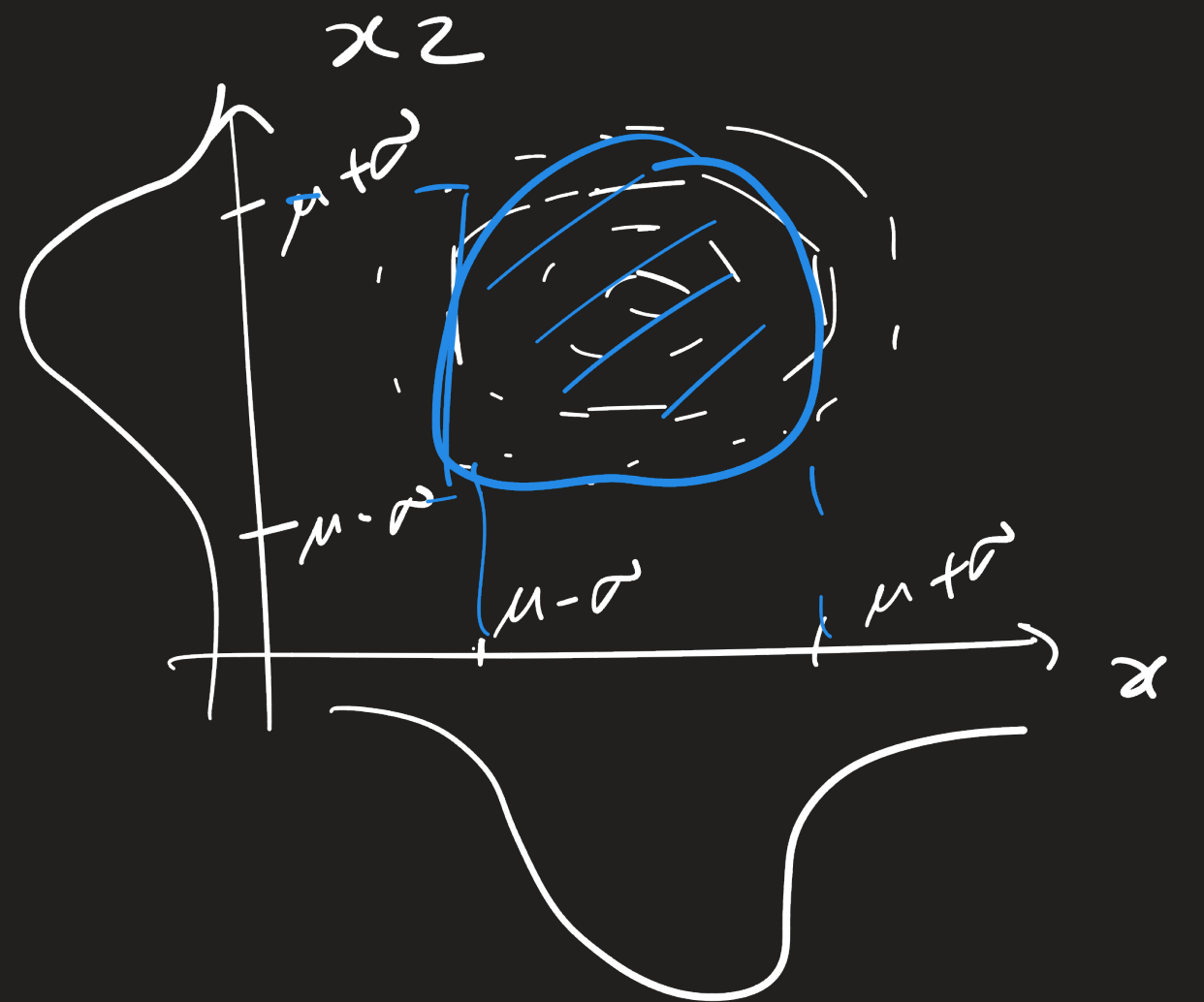
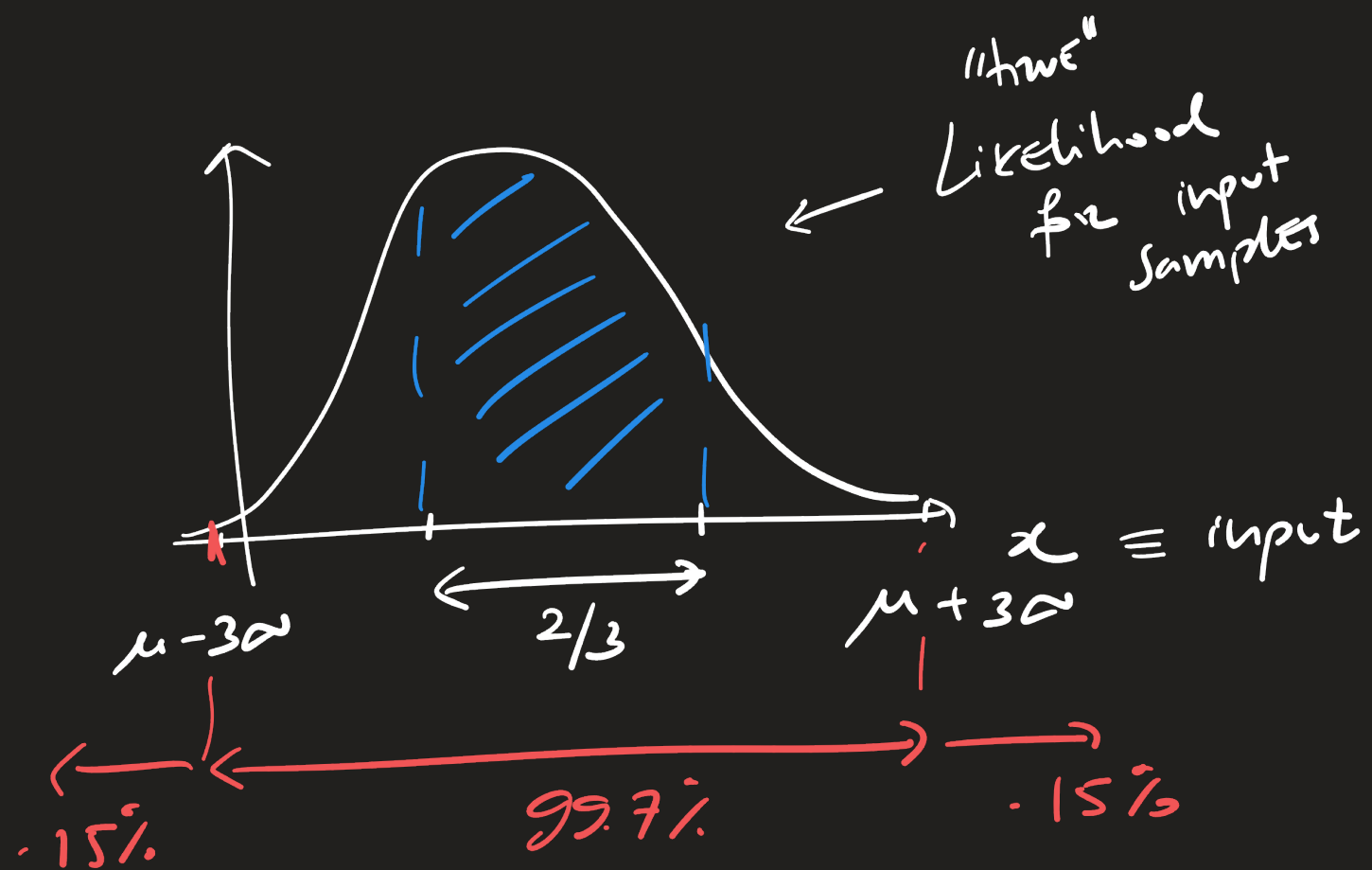
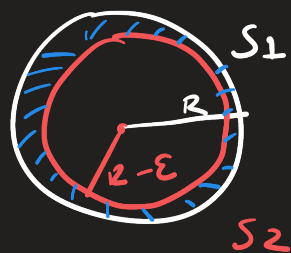






$$\epsilon \ll R$$

$V_{\text{sphere}}$   
D-dimensions  
radius R

$$= \frac{R^D \pi^{D/2}}{\Gamma(D/2 + 1)}$$

$$\frac{V_{\text{cavst}}}{V_{S1}} \xrightarrow{\text{Dimensionality} \rightarrow \infty} 1$$

$$\frac{V_{\text{cavst}}}{V_{S1}} = \frac{V_{S1} - V_{S2}}{V_{S1}} = 1 - \frac{V_{S2}}{V_{S1}}$$

$$= 1 - \frac{\frac{(R-\epsilon)^D \pi^{D/2}}{\Gamma(D/2 + 1)}}{\frac{R^D \pi^{D/2}}{\Gamma(D/2 + 1)}} = 1 - \frac{(R-\epsilon)^D}{R^D}$$

$$= 1 - \left(1 - \frac{\epsilon}{R}\right)^D \xrightarrow{D \rightarrow \infty} 1$$

# Manifold Learning

